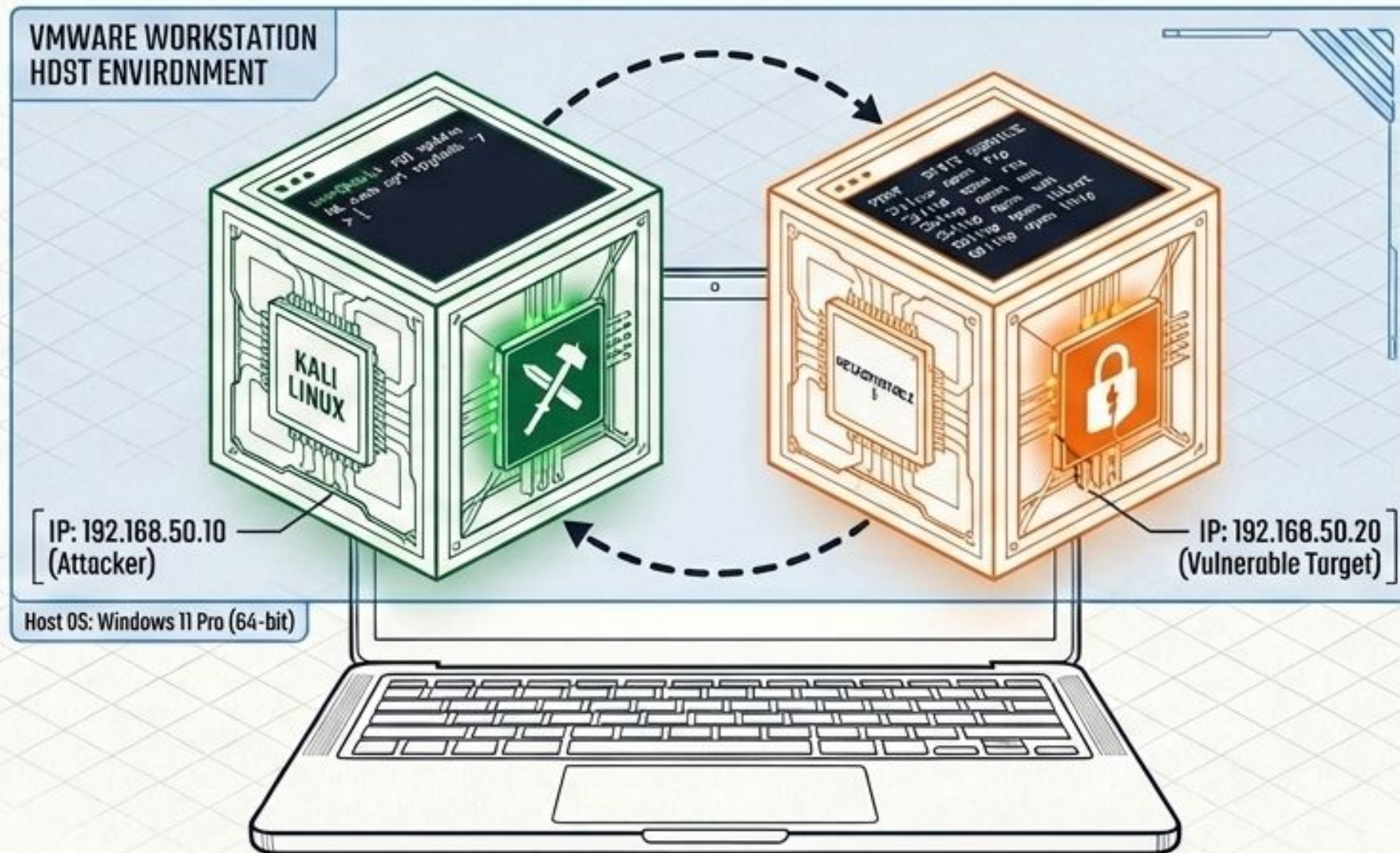


ITEC306 — ETHICAL HACKING

Lecture 3: Lab Setup — Installing Kali Linux and Metasploitable 2 on VMware Workstation.



SYSTEM REQUIREMENTS

HOST CPU:	Intel i7/AMD Ryzen 7 (Recommended)
RAM:	16GB+ (8GB Dedicated to VMs)
DISK SPACE:	100GB+ SSD

KALI LINUX SPEC CARD

CPU CORES:	2+
RAM:	4GB+
DISK:	25GB+
NETWORK:	Bridged/NAT

WARNING: EDUCATIONAL USE ONLY.
DO NOT USE ON UNAUTHORIZED NETWORKS.

1. INSTALL VMWARE
WORKSTATION

2. DOWNLOAD KALI &
METASPLOITABLE 2
ISOs

3. CREATE VIRTUAL
MACHINES

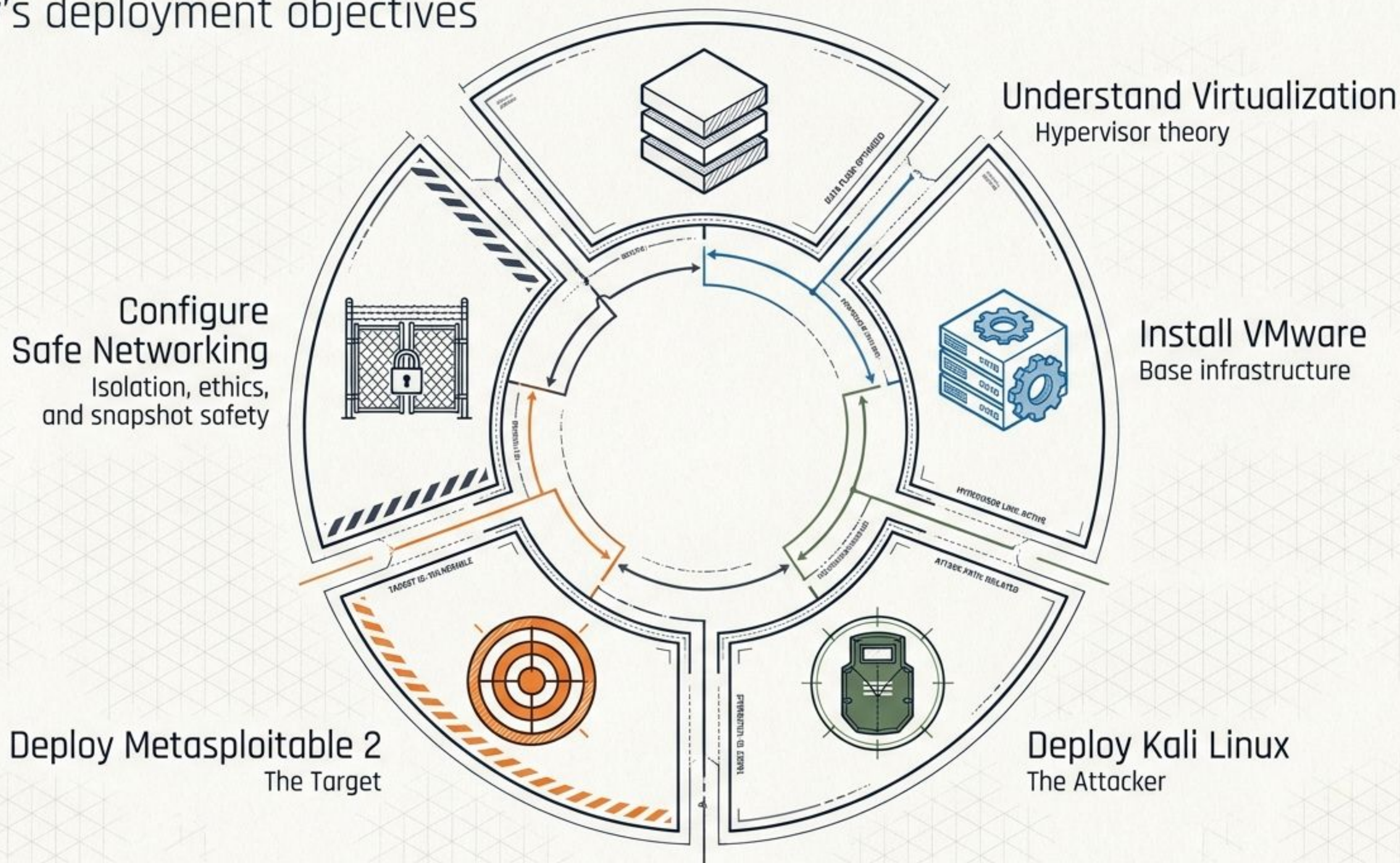
4. CONFIGURE
NETWORKING
(NAT/BRIDGED)

METASPLOITABLE 2 SPEC CARD

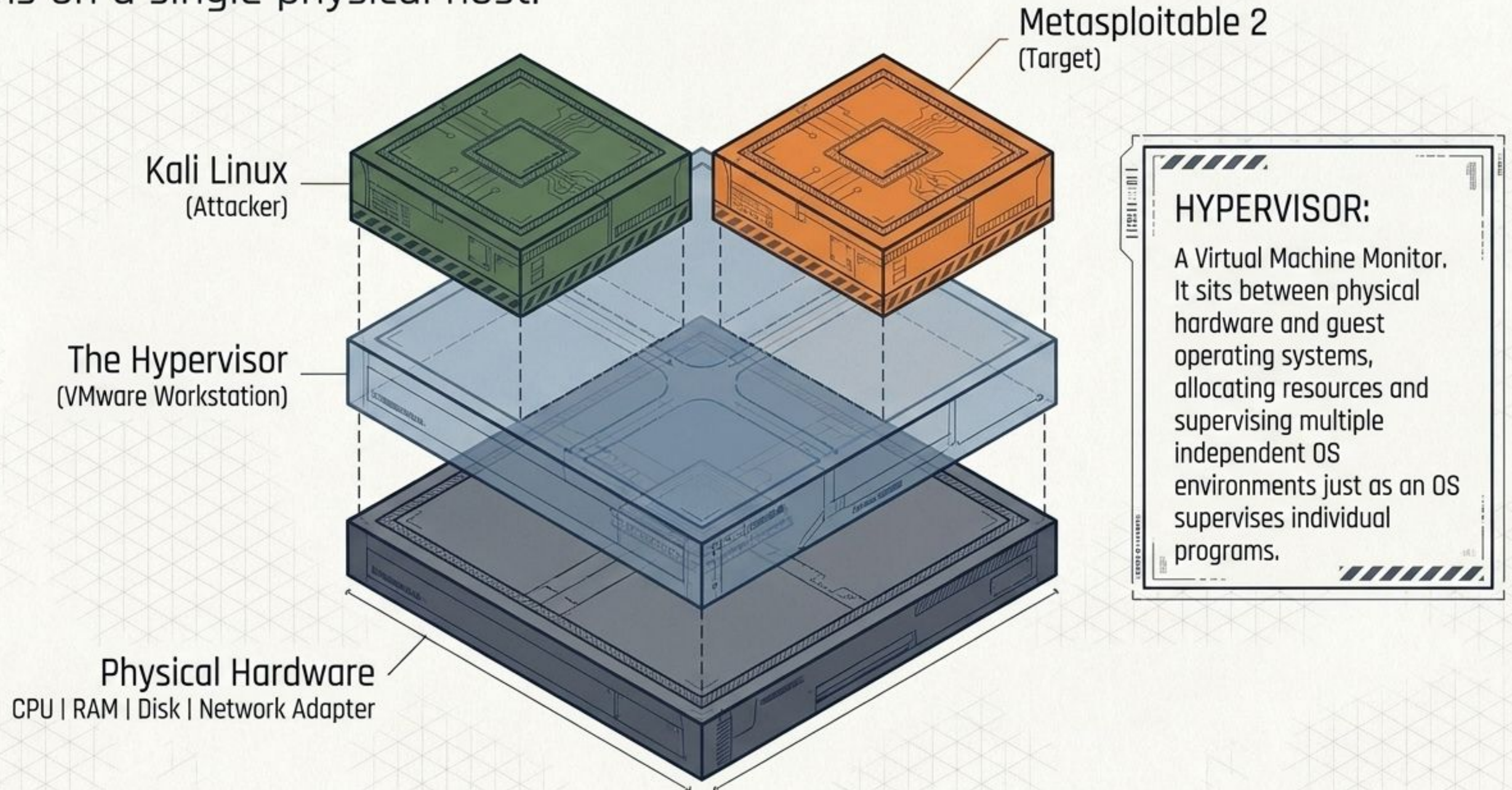
CPU CORES:	1+
RAM:	4GB+
DISK:	10GB+
NETWORK:	Bridged/NAT

WARNING: INHERENTLY VULNERABLE.
ISOLATE IN CONTAINED NETWORK.

Today's deployment objectives



Hypervisors manage isolated virtual systems on a single physical host.



Advantages and Disadvantages of Using Virtual Machines

Advantages

- Isolation and Security (Sandboxing) - Safe execution.
- Easy Recovery (Snapshots) - Quick rollback.
- Hardware Consolidation & Cost Savings - Fewer physical servers.
- Portability and Flexibility - Easily movable VMs.

Disadvantages

- Performance Overhead - Slower than bare metal.
- Resource Usage (CPU, RAM, Storage) - High system demands.
- Potential for VM Sprawl - Unmanaged VM growth.
- Licensing Costs (for some hypervisors) - Additional fees.

Types of Hypervisor

Type 1 (Bare-Metal)

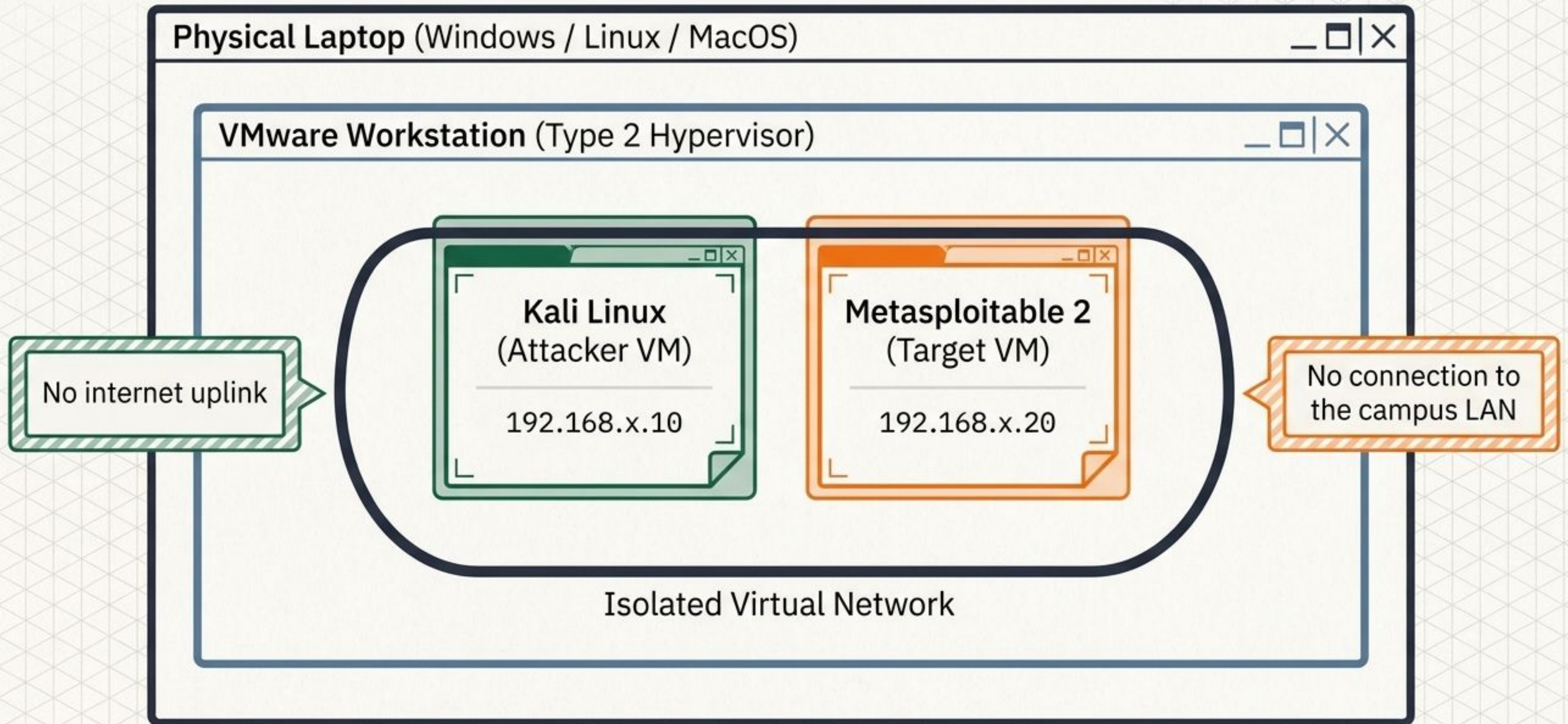
- Runs directly on physical hardware
- Installed in place of a host OS
- High performance
- Primary Use Case: Data centers & clouds
- Examples: PROXMOX, Microsoft Hyper-V

Type 2 (Hosted)

- Runs as an application on a host OS
- Installs on top of Windows / macOS / Linux
- Easiest to setup
- Primary Use Case: Student laptops
- Examples: VMware Workstation, VirtualBox

Chosen for this course: Allows safe, localized deployment on personal devices without altering the base OS.

The master blueprint for our isolated secure facility.



Provisioning the foundational infrastructure with VMware Workstation.

System Requirements



RAM
4 GB



DISK
60 GB free space required



BIOS
Intel VT-x or AMD-V enabled
(Req. for 64-bit)

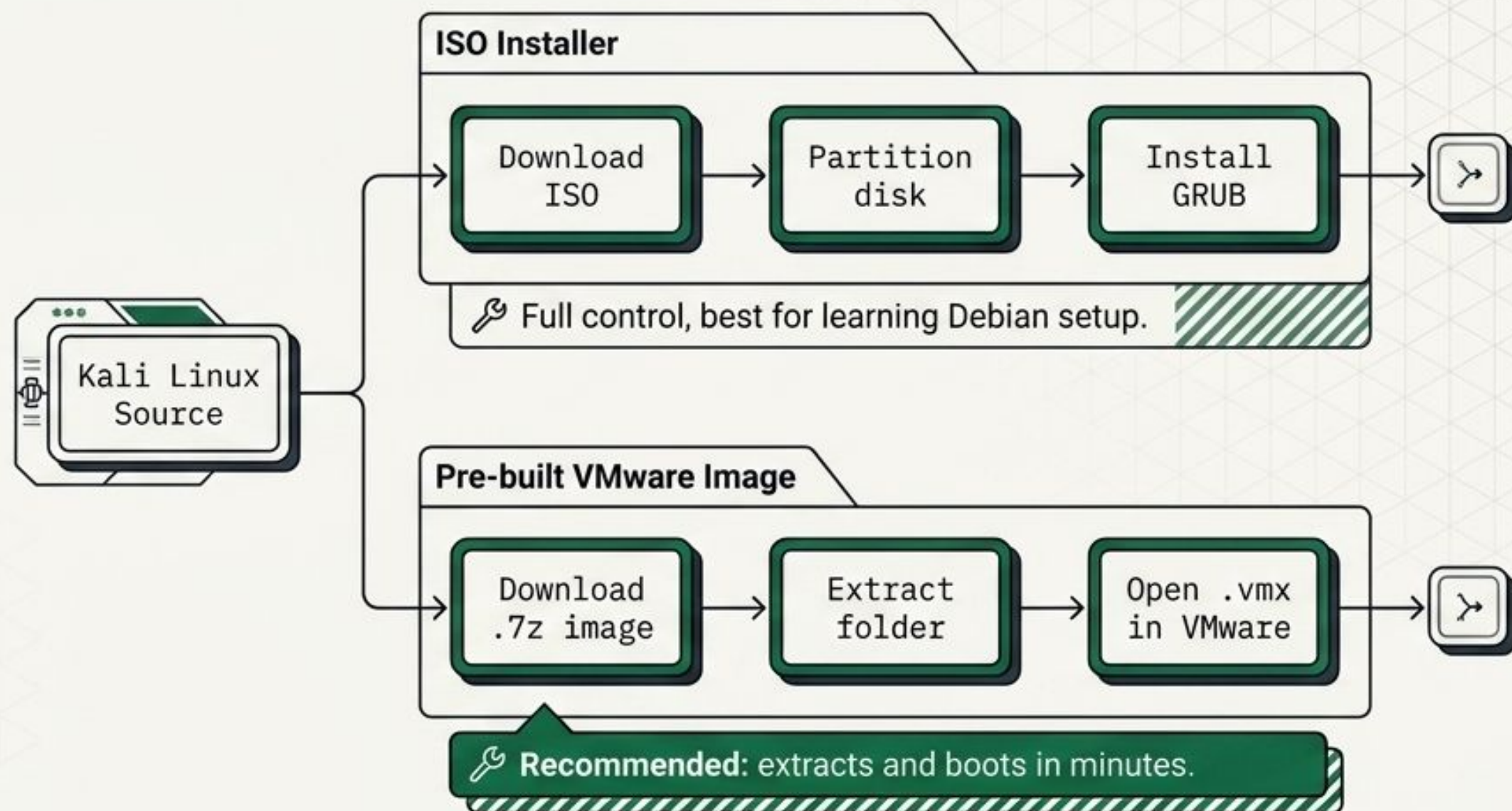
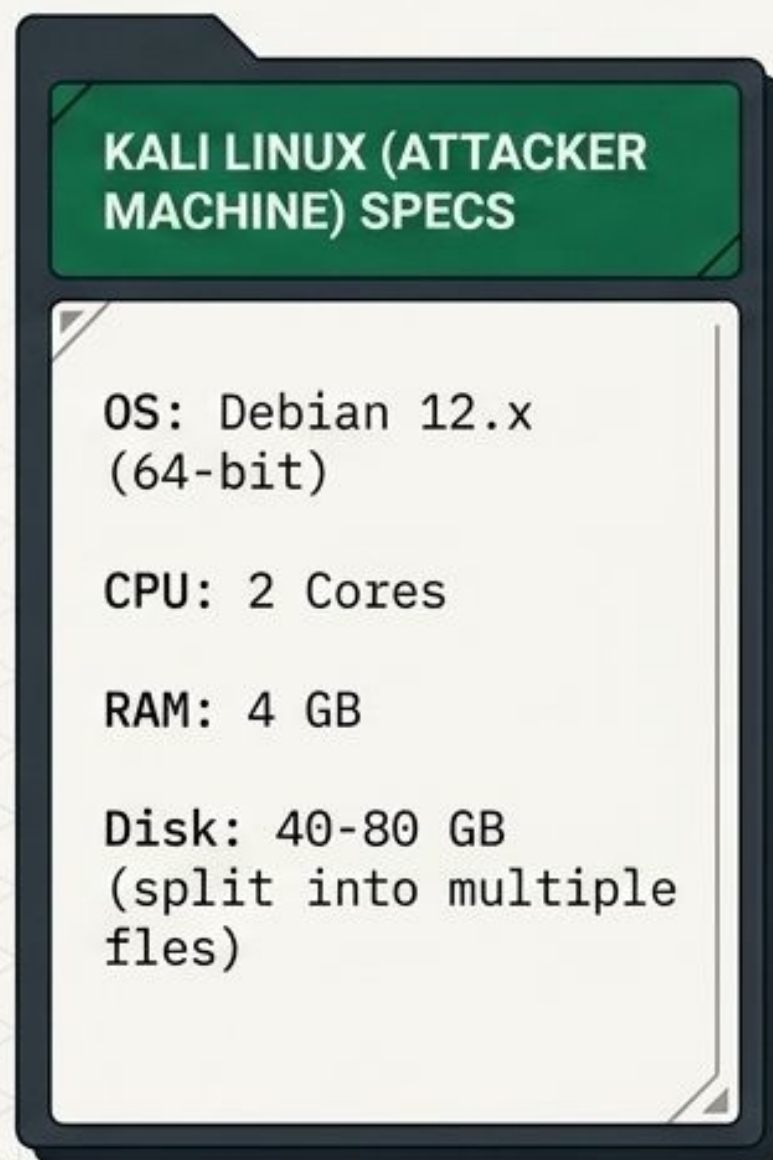
Download installer
from official Broadcom /
VMware site

Run .exe or .bundle
as Admin / Root

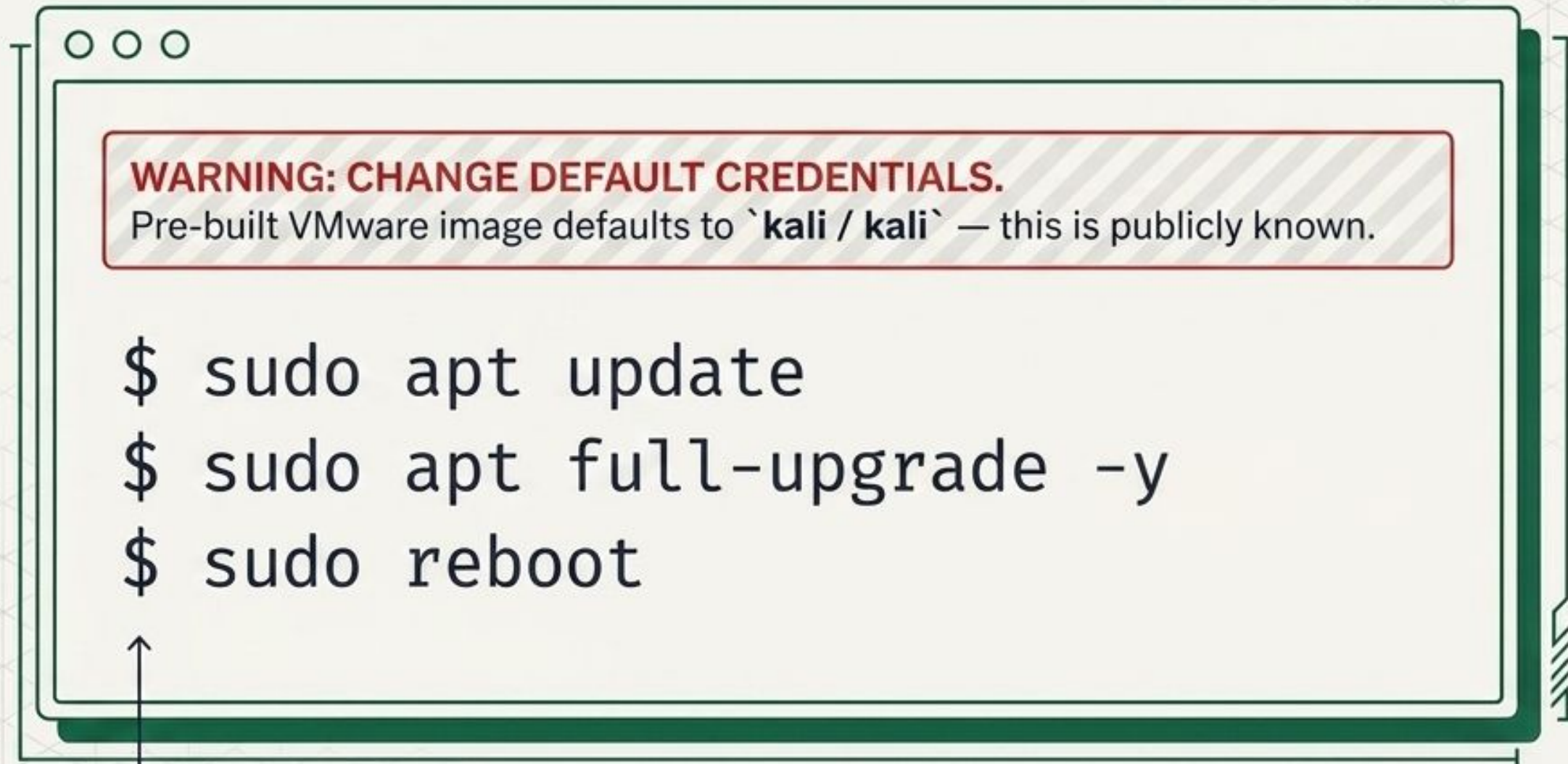
Select 'Typical' setup
in the installation wizard

Reboot physical host
before first launch

Profiling and deploying the Kali Linux attacker machine



Initializing and securing the Kali Linux environment



Confirms network access and pulls the latest vulnerability signatures before class exercises.

Deploying a deliberately vulnerable target environment



Spec Sheet

WHAT IT IS: Intentionally vulnerable Ubuntu Linux built by Rapid7.

WHY USE IT: Ships with outdated services (FTP, Samba, MySQL, Tomcat) and known backdoors for legal tool practice.

Download zip from SourceForge

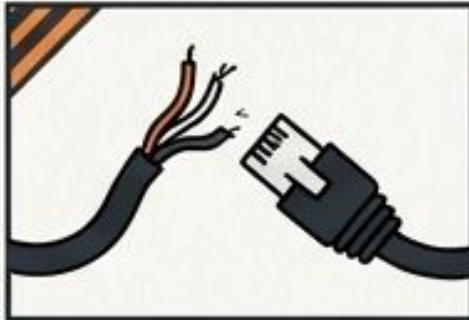
Extract to permanent folder

Open ``Metasploitable.vmx`` in VMware

Click "I Copied It" (CRITICAL: regenerates fresh network identity)

Boot directly to Ubuntu – no installer required.

Critical safety protocols for handling vulnerable software



NEVER connect Metasploitable 2 to a bridged adapter or open internet.



NEVER expose it on a production or campus network.



ONLY run on a Host-only network isolated to your physical laptop.

Treat this VM as radioactive—powerful for learning, dangerous if exposed.

Selecting a virtual network mode to guarantee isolation.

Bridged	NAT	Host-Only (VMnet)
 VM acts like a physical device on your real network. Highly dangerous for Metasploitable.	 Shares Host IP to reach the internet, hidden behind it. OK for initial Kali updates.	 VMs talk only to each other and the host. No internet path. Fully isolated, required for the lab.

Basic Linux networking commands

- **ping – Test network connectivity**

Sends ICMP packets to a host and reports response time.

```
ping 8.8.8.8
```

- **ifconfig – View/configure interfaces (legacy)**

```
$ ifconfig
```

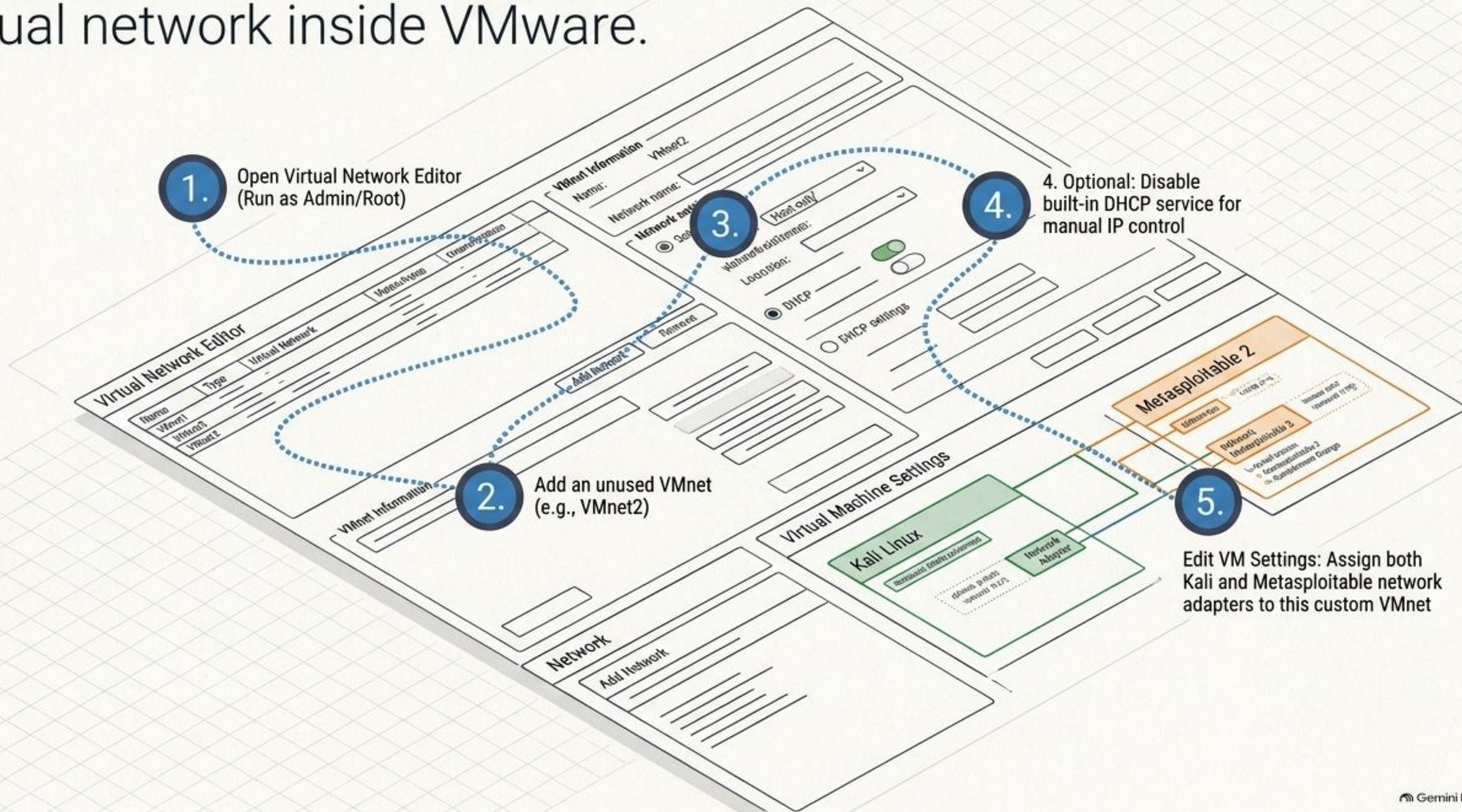
- **ip a – View IP addresses and interfaces**

Enable or disable an interface:

```
sudo ip link set [interface_name] up
```

```
sudo ip link set [interface_name] down
```


Constructing the closed-loop virtual network inside VMware.



Verifying the handshake across the isolated lab network.

```
Metasploitable 2
Default login: msfadmin / msfadmin

$ ifconfig eth0
inet addr:192.168.90.20
```

```
Kali Linux
$ ping 192.168.90.20
64 bytes from 192.168.90.20: icmp_seq=1
ttl=64 time=0.3 ms
64 bytes from 192.168.90.20: icmp_seq=2
ttl=64 time=0.2 ms
```

Success Criteria Checklist

- 0% packet loss between VMs.
- Running nmap on target shows multiple open ports (21, 22, 23, 80...).
- Zero internet access for either VM.